Compartment Fire Exposure and the Thermal Response of Unprotected Structural Steel: FDS, EN 1991-1-2 Parametric Fire and ISO 834

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Abstract

A single-compartment fire is modelled in the Fire Dynamics Simulator (FDS) and compared with the EN 1991-1-2 Annex A parametric fire and the ISO 834 standard fire. For each exposure the temperature of an unprotected steel member is calculated with the EN 1993-1-2 lumped-capacitance method, and the severity of each fire is expressed as an equivalent time of ISO 834 exposure. Two FDS fires are used: a 0.96 MW fire, and a 5 MW fire sized to resemble the parametric fire in energy and duration. For a section factor of $200\,\mathrm{m}^{-1}$, the parametric fire gives a peak steel temperature of 734 °C and an equivalent time of 20 minutes. The matched FDS fire gives 750 °C above the fire and 698 °C at the far ceiling, with equivalent times of 23 and 16 minutes, even though its peak gas temperatures are 170 K to 200 K higher. The parametric fire and the energy- and duration-matched FDS fire produce peak steel temperatures equivalent to roughly 16–23 minutes of ISO 834 exposure. These exposures subsequently cool, whereas the constant-power Run 1 fire produces a steel-temperature plateau. ISO 834 continues to rise and becomes more severe for longer exposure durations as it has no decay phase. The agreement is a consistency check, not a validation: the FDS fire was prescribed, and its energy and duration (but not its growth rate or peak) were chosen to resemble the parametric fire. Mesh size, inputs and limitations are stated throughout.

1 Introduction

Fire resistance of structural steelwork is commonly assessed using nominal temperature–time curves such as ISO 834, or natural fire models such as the parametric curves of EN 1991-1-2 Annex A. Computational fluid dynamics (CFD) can instead resolve the development and spatial variation of a compartment fire. This study applies all three descriptions to one compartment and compares the resulting steel temperatures. Some context for the comparison: an earlier draft of the Eurocode 1 parametric fire was compared with 48 experimental fire tests and correlated only weakly with measured unprotected steel temperatures [7]; uniform-temperature descriptions cannot represent fires that travel across large floor plates, for which conventional design curves cannot be assumed conservative [12]; and FDS-based natural-fire steel temperatures have previously been compared with ISO 834 for an office compartment [10].

Aim. To investigate how the representation of a compartment fire affects the predicted temperature of an unprotected steel member.

Objectives.

1. Model two compartment fires in FDS (0.96 MW, and 5 MW matched to the parametric fire in energy and duration) and extract gas temperature histories.
2. Calculate the EN 1991-1-2 Annex A parametric fire for the same compartment.
3. Calculate steel temperature with a lumped-capacitance model under ISO 834, the parametric fire and the FDS fires.
4. Express each fire as an equivalent time of ISO 834 exposure.
5. Compare the methods and state what can and cannot be concluded.

Scope. One idealised compartment and one assumed steel section factor. The study does not include a structural analysis of a member.

2 Methods

2.1 Compartment

The compartment has internal dimensions 4 m × 4 m × 2.4 m, with a single 1.0 m × 2.0 m door. The walls and ceiling are modelled as 0.2 m-thick concrete, with assumed thermal conductivity $k = 1.6\,\mathrm{W/(m\,K)}$, specific heat capacity $c = 880\,\mathrm{J/(kg\,K)}$ and density $\rho = 2300\,\mathrm{kg/m^3}$. These properties give a thermal inertia of

$$b = \sqrt{k\rho c} \approx 1800\,\mathrm{J/(m^2 s^{1/2} K)},$$

which is used in the EN 1991-1-2 Annex A parametric-fire calculation.

2.2 Standard fire (ISO 834)

$$\theta_g = 20 + 345\log_{10}(8t + 1) \tag{1}$$

with t in minutes. This is the standard temperature–time curve of EN 1991-1-2, which is the ISO 834 curve [1, 3]. It rises continuously and has no cooling phase.

2.3 EN 1991-1-2 Annex A parametric fire

The heating phase is calculated using the parametric temperature–time relationship in EN 1991-1-2, Annex A [1, Annex A, p. 30]:

$$\theta_g = 20 + 1325\left(1 - 0.324e^{-0.2t^*} - 0.204e^{-1.7t^*} - 0.472e^{-19t^*}\right), \qquad t^* = \Gamma t \tag{2}$$

where t is in hours and

$$\Gamma = \frac{(O/b)^2}{(0.04/1160)^2}, \qquad O = \frac{A_v\sqrt{h_{eq}}}{A_t}. \tag{3}$$

The time to reach the maximum gas temperature is calculated as $t_{\max} = 0.2 \times 10^{-3} q_{t,d}/O$, with $q_{t,d}$ in $\mathrm{MJ/m^2}$ and O in $\mathrm{m^{1/2}}$. For this case, $t_{\max} = 34.7\,\mathrm{min}$, exceeding the limiting time $t_{\lim} = 20\,\mathrm{min}$. The fire is therefore ventilation-controlled. The cooling phase is calculated according to Annex A, and the parametric-fire inputs are summarised in Table 1.

Table 1: Parametric fire inputs and results.

Parameter	Value	Unit
Floor area A_f	16.0	$\mathrm{m^2}$
Total enclosure area A_t (incl. openings)	70.4	$\mathrm{m^2}$
Opening factor O	0.0402	$\mathrm{m^{1/2}}$
Thermal inertia b	1800	$\mathrm{J/(m^2 s^{1/2} K)}$
Fire load per floor area $q_{f,d}$ (office, 80 % fractile)	511	$\mathrm{MJ/m^2}$
Fire load per enclosure area $q_{t,d}$	116.1	$\mathrm{MJ/m^2}$
Γ	0.419	–
t_{lim} (medium growth, Annex A clause (10))	20	min
t_{max}	34.7	min
Peak gas temperature	751	°C

The value $511\,\mathrm{MJ/m^2}$ is the characteristic 80 % fractile fire load for offices (EN 1991-1-2 Table E.4). It was used directly as the design value $q_{f,d}$, i.e. the combustion factor m and the factors δ_{q1}, δ_{q2}, δ_n of Annex E (E.1) were taken as 1.0. Applying $m = 0.8$ and $\delta_{q1} = 1.10$ (the value for the smallest floor area in Table E.1, $25\,\mathrm{m^2}$; this room is smaller) would give about $450\,\mathrm{MJ/m^2}$, so the value used is about 12 % higher. Offices are classed as medium fire growth ($t_\alpha = 300\,\mathrm{s}$, $\mathrm{RHR}_f = 250\,\mathrm{kW/m^2}$, Table E.5) [1, Annex E, Tables E.1, E.4 and E.5].

2.4 FDS model

Geometry and fires. The room is as above. Two fires were modelled, both with propane as the fuel (soot yield 0.01). *Run 1* used a 0.8 m × 0.8 m burner (0.64 m^2) at 1500 kW/m^2, giving 0.96 MW. The output grew as t^2, reached full power at 150 s and stayed constant for the whole 2400 s run. *Run 2* (the energy- and duration-matched fire) used a 2 m × 2 m burner at 1250 kW/m^2, giving 5 MW. Its output grew as t^2 to full power at 300 s, stayed constant to 1500 s, then decreased linearly to zero at 2100 s. The run lasted 2400 s. The total energy of Run 2 was set to about 8000 MJ, close to the 8176 MJ ($q_{f,d}A_f$) assumed by the parametric fire, and its 35 min burning time is close to the parametric t_{max} of 34.7 min. Run 2 is not the Annex E fire: the Annex E medium growth curve $Q = 10^6(t/300)^2$ W reaches 1 MW at 300 s and is capped at $\mathrm{RHR}_f A_f = 4$ MW, whereas Run 2 reaches 5 MW at 300 s. The ventilation-controlled limit of Annex E (E.6), $0.10\, m\, H_u A_v \sqrt{h_{eq}}$ with $m = 0.8$, $H_u = 17.5$ MJ/kg, $A_v = 2.0$ m^2 and $h_{eq} = 2.0$ m, is about 4.0 MW, so Run 2 exceeds it and some burning outside the door is expected. Annex E also starts the decay when 70 % of the fire load has burned, whereas Run 2 starts it at about 80 %. Annex A does not explicitly model fuel chemistry; its parametric curve is determined by the fire load, ventilation and enclosure thermal properties. Matching the prescribed FDS fire to the parametric fire in energy and duration therefore provides a useful comparison, but does not make the two fire exposures equivalent. The domain extends beyond the room so that flames and hot gas leaving the door are included.

Mesh Resolution. Simulations were run with FDS version 6.11.1 and results were visualised in Smokeview 6.11.2. Both simulations used a single uniform mesh with a nominal cell width of 0.2 m. Run 1 used $30 \times 35 \times 15 = 15{,}750$ cells, while Run 2 used $30 \times 40 \times 30 = 36{,}000$ cells; the latter domain was taller and extended further in front of the door to accommodate potential flame spill. The characteristic fire diameter [5, Section 6.3.6, p. 41] was calculated as

$$D^* = \left(\frac{\dot{Q}}{\rho_\infty c_p T_\infty \sqrt{g}} \right)^{2/5}$$

using standard ambient conditions. At the peak heat release rates of 0.96 MW and 5 MW, D^* was approximately 0.94 m and 1.83 m, respectively, corresponding to $D^*/\delta x$ ratios of 4.7 and 9.2 for the 0.2 m mesh. These ratios provide an indication of mesh resolution but do not independently establish numerical accuracy. The FDS User Guide recommends starting with a relatively coarse mesh and refining it until no appreciable differences are seen in the results (a mesh sensitivity study), and uses $D^*/\delta x$ as the measure of resolution for buoyant plumes [5, Section 6.3.6, p. 41]. A supplementary Run 1 simulation using a refined 0.1 m mesh was performed for the first 60 s. Gas temperatures at the monitored locations reportedly agreed with those from the 0.2 m simulation to within approximately 10%. However, the heat release rate at 60 s was only approximately 0.154 MW, so this comparison assesses early fire growth rather than the peak-fire condition. No equivalent mesh-sensitivity study was completed for Run 2, and the full-duration refined simulation was not practical with the available computational resources. Consequently, the results should be interpreted with these mesh-resolution limitations in mind.

Devices. Thermocouples were placed under the ceiling above the fire (2.3 m high; at $y = 2.0$ m for Run 1 and 2.5 m for Run 2), under the ceiling away from the fire (3.5 m, 3.5 m, 2.3 m), and at heights of 1.9, 1.5, 1.1 and 0.5 m above the floor at the second position. A thermocouple reads a temperature that differs from the true gas temperature, because the bead exchanges radiation with the walls, hot gases and soot, and the surroundings [9, p. 4], and it responds to changes with some lag. Adiabatic surface temperature (AST), the temperature of a surface that can neither absorb nor lose heat to its surroundings [8, p. 1001], was recorded in a 60 s test only, where AST was lower than the thermocouple early on because the surrounding surfaces are cold. Run 2 also

recorded AST, but the steel calculations use the thermocouple temperatures for consistency with Run 1.

2.5 Steel temperature

An unprotected steel member is assumed, with a section factor $A_m/V = 200\,\mathrm{m}^{-1}$. This is representative of a light open section and is an assumed value rather than one associated with a specific universal beam (UB). The steel temperature is calculated using the lumped-capacitance method in EN 1993-1-2 [2, Clause 4.2.5.1, p. 37]. The temperature increment over a time step Δt is

$$\Delta\theta_a = k_{sh}\frac{A_m/V}{c_a\rho_a}\,\dot{q}_{net}\,\Delta t,$$

where the net heat flux is

$$\dot{q}_{net} = \alpha_c(\theta_g - \theta_a) + \Phi\varepsilon_m\varepsilon_f\sigma\left(T_g^4 - T_a^4\right).$$

Temperatures are expressed in kelvin in the radiation term. The parameters are $k_{sh} = 1$ (no section-factor reduction applied), $\Phi = 1$ and $\varepsilon_f = 1.0$ [1, Clause 3.1], $\varepsilon_m = 0.7$ [2], and $\rho_a = 7850\,\mathrm{kg/m}^3$. The temperature-dependent specific heat $c_a(\theta)$ is taken from EN 1993-1-2, Clause 3.4.1.2 [2, Clause 3.4.1.2, equations (3.2a)–(3.2d)]. The convective heat-transfer coefficient was set to $\alpha_c = 25\,\mathrm{W/(m^2K)}$ for ISO 834 [1, Clause 3.2.1, p.24] and $35\,\mathrm{W/(m^2K)}$ for the natural-fire simulations, the value EN 1991-1-2 gives for natural fire models [1, Section 3.3, p.25]. Applying it to the FDS thermocouple temperatures is an assumption of this study. A time step of 1 s is used for all fire exposures. The calculation was implemented independently in Python and MATLAB, with matching peak steel temperatures.

2.6 Equivalent time

The equivalent time t_e is defined as the duration of exposure to the ISO 834 standard temperature–time curve that produces the same peak steel temperature as the fire being assessed. It is found by running the ISO 834 exposure through the same steel-temperature model until the corresponding peak steel temperature is reached. This is a temperature-based application of the equivalent time of fire exposure concept defined in EN 1991-1-2 [1, Section 1.5.1.1, p.11], and relating a natural fire to the standard test through the time a steel member takes to reach a given temperature was developed by Law [6]. It differs from the equivalent time in EN 1991-1-2 Annex F, which is calculated from the design fire load density, a ventilation factor and enclosure and material factors rather than from a steel temperature [1, Annex F, p. 53]. The resulting equivalent time is specific to the assumed section factor, steel thermal properties and heat-transfer model, and does not imply equivalence in other aspects of the fire exposure.

3 Results

3.1 ISO 834 and the parametric fire

The parametric gas temperature peaked at 751 °C at 34.7 min. ISO 834 passes 751 °C after about 16 min and keeps rising, reaching 1110 °C at 3 h. For $A_m/V = 200\,\mathrm{m}^{-1}$ the steel under the parametric fire peaked at 734 °C at 38.5 min. Under ISO 834 the steel is still heating after 3 h (1109 °C), so ISO 834 has no meaningful peak.

3.2 FDS fires

Run 1 reached its 0.96 MW plateau at 150 s. Run 2 followed the prescribed curve (Figure 1): the heat release rate reached about 5 MW at 300 s, fluctuated between roughly 3.5 and 5.9 MW (mean about 5 MW) to 1500 s, and fell to zero at 2100 s. The fuel mass flow was steady at 0.109 kg/s, equivalent to about 5 MW, so FDS did not reduce the burning. Integrating the record gives 7983.5 MJ.

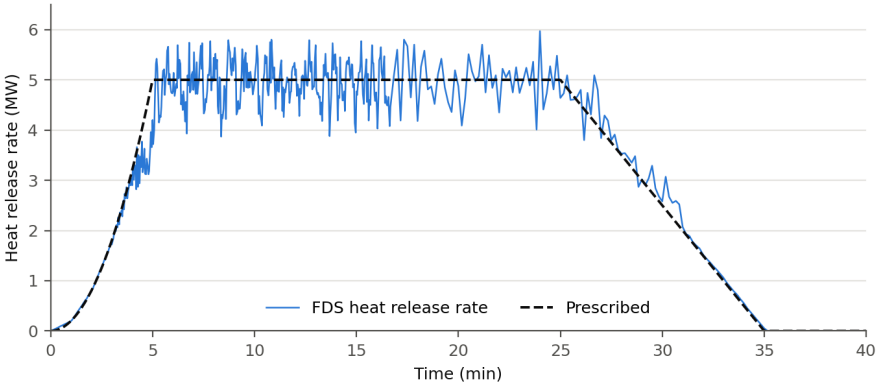


Figure 1: Heat release rate in Run 2 (energy- and duration-matched fire).

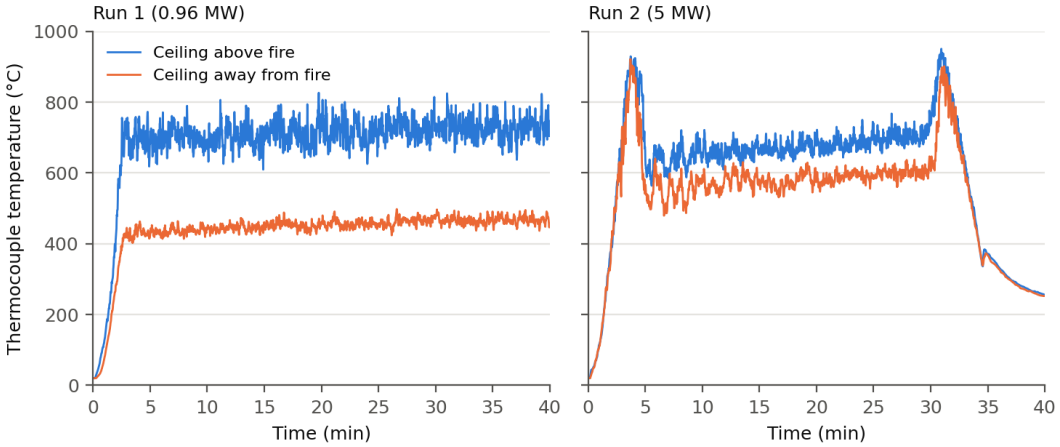


Figure 2: FDS thermocouple temperatures, Run 1 and Run 2 (0.2 m mesh).

In Run 1 the gas under the ceiling above the fire peaked at 826 °C, and at the far ceiling at 498 °C. In Run 2 the peaks were 950 °C and 922 °C. Temperatures fell toward the floor in both runs, showing a hot upper layer. In Run 2 a temperature slice through the doorway showed gas outside the opening reaching about 800 °C at its hottest, never hotter than the gas inside, consistent with the upper layer spilling out of the door. Some combustion outside the opening cannot be ruled out and was not quantified. Venting flames of this kind have been compared between CFD and Eurocode correlations elsewhere [11], but that was not attempted here.

3.3 Steel temperature and equivalent time

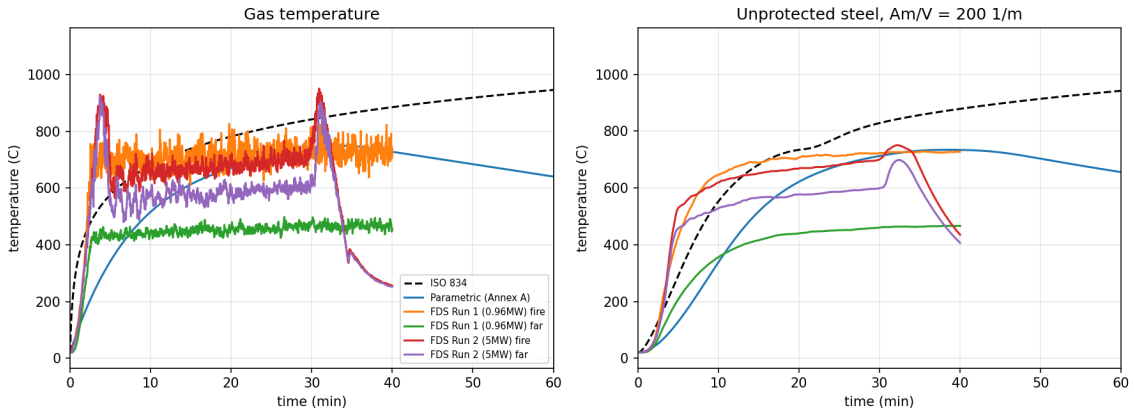


Figure 3: Gas temperature (left) and unprotected steel temperature (right) for $A_m/V = 200\,\mathrm{m}^{-1}$ under ISO 834, the parametric fire and the FDS fires.

Table 2: Peak gas temperature, peak steel temperature, equivalent time t_e and $k_{y,\theta}$ at the peak steel temperature, for $A_m/V = 200\,\mathrm{m}^{-1}$. $k_{y,\theta}$ is interpolated linearly from EN 1993-1-2 Table 3.1, which the standard permits for intermediate temperatures. Peak gas is the peak thermocouple reading.

Exposure	Peak gas (°C)	Peak steel (°C)	t_e (min)	$k_{y,\theta}$
Parametric (Annex A)	751	734	20.0	0.19
FDS Run 1 (0.96 MW), above fire	826	728	18.9	0.20
FDS Run 1 (0.96 MW), far ceiling	498	467	8.0	0.85
FDS Run 2 (5 MW), above fire	950	750	22.6	0.17
FDS Run 2 (5 MW), far ceiling	922	698	16.1	0.23

The Run 1 steel temperature above the fire (728 °C) is close to the parametric value, but it is not a peak. The 0.96 MW fire never decays, so the steel reaches a plateau of about 728 °C at about 35 min, shortly before the run ends at 40 min.

3.4 Effect of section factor

Table 3 gives peak steel temperature and equivalent time for three section factors. The ranking of the exposures is the same in every case.

Table 3: Peak steel temperature (°C) and t_e (min) for three section factors.

	A_m/V (m^{-1})		
Exposure	100	200	300
Parametric (Annex A)	723 / 23.6	734 / 20.0	737 / 19.0
FDS Run 1, above fire	723 / 23.6	728 / 18.9	730 / 17.5
FDS Run 1, far ceiling	457 / 11.7	467 / 8.0	468 / 6.5
FDS Run 2, above fire	729 / 24.4	750 / 22.6	781 / 23.1
FDS Run 2, far ceiling	650 / 18.6	698 / 16.1	722 / 16.4

3.5 Strength reduction

At the peak temperatures above, the effective yield strength factor $k_{y,\theta}$ is between 0.17 and 0.23 for the parametric fire and Run 2, and 0.85 at the far ceiling in Run 1 (Table 2). These factors apply to the steel material only. Member resistance depends on the applied load ratio, which is not considered here.

4 Discussion

Natural fires against ISO 834. Every natural-fire exposure gave a t_e between about 6 and 24 minutes across the three section factors, against a standard fire that keeps rising. For the parametric fire, t_e is about 20 minutes for $A_m/V = 200\,\mathrm{m}^{-1}$. The natural fire is therefore far less severe than the 60 minute ISO 834 exposure often used for a rating. ISO 834 does not necessarily represent the more severe thermal exposure in every compartment-fire scenario. Experimental fires exhibit a range of temperature–time histories, with peak temperatures potentially exceeding the standard curve before cooling [4, Fig. 3.19, Section 3.7.2, p. 61]. The relative severity of a natural fire and the ISO 834 exposure depends on the fire scenario and compartment characteristics. Unlike ISO 834, the parametric fire accounts for factors including ventilation, enclosure thermal properties and fire load [1, Annex A, p. 30]. This is why natural-fire methods are used where the cost of protection justifies the effort. Szumigała and Polus [10, Fig. 13, p. 169] similarly found that FDS natural-fire modelling gave different unprotected steel temperatures from the ISO 834 curve for an office compartment.

Annex A and FDS for a comparable fire. Run 2 was sized to resemble the parametric fire in energy and duration, and the two methods agree closely on steel temperature: peak steel 750 °C above the fire and 698 °C at the far ceiling, against 734 °C for Annex A. At $A_m/V = 200\,\mathrm{m}^{-1}$ the differences are +16 and −36 K, and across the three section factors from about −73 to 44 K. This holds although the FDS peak gas temperatures (922–950 °C) exceed the parametric peak of 751 °C by about 170–200 K. Steel temperature depends on the heat transferred to the member throughout the fire exposure, rather than on peak gas temperature alone. The Eurocode thermal-response model calculates the temperature increase over successive time intervals as a function of net heat flux, section factor and the thermal properties of steel [2, Clause 4.2.5.1, p. 37]. Consequently, the higher but shorter-lived FDS gas-temperature peak can produce a similar peak steel temperature to the lower, longer parametric exposure.

Spatial variation. Annex A assumes a uniform gas temperature throughout the compartment, whereas FDS resolves spatial variations in temperature [4, Clauses 11.2.3–11.2.4, p. 343]. In large open-plan compartments the non-uniformity can be much greater, because the fire may travel across the floor plate, and design fires that ignore this cannot be assumed conservative, as shown for a concrete frame [12, p. 1635]. A 4 m × 4 m room is not that case. In Run 2, the peak steel temperature above the fire was approximately 750 °C, compared with 698 °C at the far ceiling. The difference was considerably greater in Run 1, at 728 °C and 467 °C, respectively. In Run 2, the Annex A prediction of 734 °C lies between the two FDS results, slightly underestimating the peak above the fire while overestimating the temperature at the far ceiling.

What the agreement does and does not mean. FDS does not predict the fire here. The heat release rate was prescribed, and the energy and duration of Run 2 were chosen to resemble the parametric fire, so close agreement is partly built in. Both methods also depend on assumed fire loads. The result shows that the two methods are consistent for a simple, ventilated compartment within the limits of Annex A. It does not show that either is correct, and no experimental data were used. Franssen [7, p. 938] compared an earlier draft of the Eurocode 1 parametric fire (its

Annex B) with 48 experimental fire tests. Predicted maximum gas temperature did not correlate with the measurements, unprotected steel temperature correlated only weakly, and protected steel temperature correlated better but was almost systematically under-predicted. That is a result for the earlier draft, not for the EN 1991-1-2 Annex A used here, but it illustrates why agreement between two models is not evidence that either matches real fires.

When is the CFD worth the cost? For this compartment, which lies within the limits of Annex A, FDS produced peak steel temperatures broadly comparable to the simplified method. The main reduction in predicted steel temperature came from using a natural-fire exposure rather than ISO 834, which Annex A also provides. The Annex A calculation took approximately 2.05 s to execute, compared with runtimes of several hours for the FDS cases. The additional spatial detail provided by CFD may be valuable where variations in fire exposure affect the structural assessment. For the simple compartment considered here, however, the additional modelling effort is unlikely to be justified by peak steel temperature alone.

4.1 Limitations

- Idealised geometry, a single compartment, and prescribed fire sources.
- Run 2 was sized to resemble the parametric fire in energy and duration, so agreement is not independent of the choice of fire.
- Run 2 is not the Annex E medium-growth fire: it grows faster (5 MW at 300 s against 1 MW) and peaks higher (5 MW against about 4 MW from both $\mathrm{RHR}_f A_f$ and the ventilation limit). The effect on the steel temperature was not tested.
- The fire load 511 MJ/m^2 is the 80 % fractile characteristic value used as the design value with m and the δ factors taken as 1.0; applying $m = 0.8$ and $\delta_{q1} = 1.10$ would give about 450 MJ/m^2.
- Some of Run 2 may have burned outside the door; the fraction was not quantified.
- Coarse mesh ($\delta x = 0.2$ m), with a 60 s check only at 0.1 m (Run 1, at about 0.15 MW); none for Run 2.
- Thermocouple temperatures are not true gas or steel-surface temperatures. Above the 5 MW fire the flame is likely to reach the ceiling, so the thermocouple reading there does not capture the flame heat flux to a member.
- Run 1 ends at 40 min with the steel still on a plateau, so its steel "peak" is a steady value.
- Assumed A_m/V, α_c, ε_m; no shadow effect.
- Lumped capacitance ignores temperature gradients in the section.
- No experimental validation of any model.
- Annex A assumes a mainly cellulosic fire load, whereas the FDS fires burn propane. This affects the heat release and soot production in FDS but not the Annex A curve, which depends only on fire load and ventilation.
- The Annex A validity limits were checked for this case: floor area up to 500 m^2, height up to 4 m, no roof openings, O 0.02–0.20, b 100–2200 and $q_{t,d}$ 50–1000. The method also assumes the fire load burns out completely. The room is small and the method is approximate.

5 Conclusions

1. For this compartment and $A_m/V = 200\,\mathrm{m}^{-1}$, the Annex A parametric fire heats unprotected steel to 734 °C, equivalent to about 20 minutes of ISO 834 exposure. All natural-fire exposures gave t_e between about 6 and 24 minutes over the three section factors studied.
2. An FDS fire sized to resemble the parametric fire in energy and duration gave peak steel temperatures of 750 °C above the fire and 698 °C at the far ceiling, within about +16 and −36 K of Annex A, despite peak gas temperatures 170 K to 200 K higher.

3. FDS showed a spatial variation in steel temperature that Annex A cannot represent: about 50 K between the near and far ceiling in the matched run.
4. The agreement is a consistency check for a simple compartment within the limits of Annex A and not a validation, because the FDS fire was prescribed and sized to resemble the parametric fire.
5. For a simple compartment of this kind, the main benefit comes from using a natural fire instead of ISO 834. CFD adds spatial information at much greater cost, and is likely to be justified where the assumptions of Annex A fail.

References

[1] BSI, *BS EN 1991-1-2: Eurocode 1 – Actions on structures – Part 1-2: Actions on structures exposed to fire*. London: British Standards Institution, 2002.

[2] BSI, *BS EN 1993-1-2:2005, Eurocode 3: Design of steel structures – Part 1-2: General rules – Structural fire design*. London: British Standards Institution, 2005.

[3] ISO, *ISO 834-1: Fire-resistance tests – Elements of building construction – Part 1: General requirements*.

[4] A. H. Buchanan and A. K. Abu, *Structural Design for Fire Safety*, 2nd ed., Wiley, 2017.

[5] K. McGrattan, R. McDermott, C. Weinschenk and G. Forney, *Fire Dynamics Simulator User's Guide*, Sixth Edition, NIST Special Publication 1019, National Institute of Standards and Technology, Gaithersburg, MD, 2013. doi:10.6028/NIST.sp.1019

[6] M. Law, *A relationship between fire grading and building design and contents*, Fire Research Note No. 877, Joint Fire Research Organization, Borehamwood, 1971.

[7] J.-M. Franssen, Improvement of the parametric fire of Eurocode 1 based on experimental test results, in *Fire Safety Science: Proceedings of the Sixth International Symposium*, International Association for Fire Safety Science, 2000, pp. 927–938. doi:10.3801/IAFSS.FSS.6-927

[8] U. Wickström, The adiabatic surface temperature and the plate thermometer, in *Fire Safety Science: Proceedings of the Tenth International Symposium*, International Association for Fire Safety Science, 2011, pp. 1001–1011. doi:10.3801/IAFSS.FSS.10-1001

[9] L. G. Blevins and W. M. Pitts, *Modeling of Bare and Aspirated Thermocouples in Compartment Fires*, NISTIR 6310, National Institute of Standards and Technology, Gaithersburg, MD, 1999. doi:10.6028/NIST.IR.6310

[10] M. Szumigała and Ł. Polus, A comparison of the rise of the temperature of an unprotected steel column subjected to the standard fire curve ISO 834 and to a natural fire model in the office, *Engineering Transactions*, vol. 63, no. 2, pp. 157–170, 2015. doi:10.24423/engtrans.59.2015

[11] E. Asimakopoulou, D. Kolaitis and M. Founti, Comparative assessment of CFD tools and the Eurocode methodology in describing externally venting flames, *MATEC Web of Conferences*, vol. 9, article 03003, 2013. doi:10.1051/matecconf/20130903003

[12] A. Law, J. Stern-Gottfried, M. Gillie and G. Rein, The influence of travelling fires on a concrete frame, *Engineering Structures*, vol. 33, pp. 1635–1642, 2011.

A FDS Input Files

This appendix presents the FDS input files used to define the compartment geometry, fire scenarios, material properties and temperature measurement locations.

A.1 Run 1: Baseline Fire (0.96 MW)

```
&HEAD CHID='compartment1', TITLE='Compartment fire, 4x4x2.4 m room, 1.0x2.0 m door, 0.2 m mesh' /

&MESH IJK=30,35,15, XB=-1.0,5.0,-2.0,5.0,0,3.0 /
&TIME T_END=2400.0 /
&DUMP DT_DEVC=1.0, DT_SLCF=2.0 /

&REAC FUEL='PROPANE', SOOT_YIELD=0.01 /

&MATL ID='CONCRETE', CONDUCTIVITY=1.6, SPECIFIC_HEAT=0.88, DENSITY=2300.0 /
&SURF ID='CONCRETE', MATL_ID='CONCRETE', THICKNESS=0.2, COLOR='GRAY' /

&SURF ID='BURNER', HRRPUA=1500.0, TAU_Q=-150.0, COLOR='RED' /

&OBST XB=-0.2,4.2,-0.2,0.0,0,2.6, SURF_ID='CONCRETE' /
&OBST XB=-0.2,4.2,4.0,4.2,0,2.6, SURF_ID='CONCRETE' /
&OBST XB=-0.2,0.0,-0.2,4.2,0,2.6, SURF_ID='CONCRETE' /
&OBST XB=4.0,4.2,-0.2,4.2,0,2.6, SURF_ID='CONCRETE' /
&OBST XB=-0.2,4.2,-0.2,4.2,2.4,2.6, SURF_ID='CONCRETE' /

&HOLE XB=1.4,2.4,-0.3,0.1,0,2.0 /

&OBST XB=1.6,2.4,1.6,2.4,0,0.2, SURF_IDS='BURNER','INERT','INERT' /

&VENT MB='XMIN', SURF_ID='OPEN' /
&VENT MB='XMAX', SURF_ID='OPEN' /
&VENT MB='YMIN', SURF_ID='OPEN' /
&VENT MB='YMAX', SURF_ID='OPEN' /
&VENT MB='ZMAX', SURF_ID='OPEN' /

&SLCF PBY=2.0, QUANTITY='TEMPERATURE' /
&SLCF PBX=1.9, QUANTITY='TEMPERATURE' /
&SLCF PBY=2.0, QUANTITY='VELOCITY', VECTOR=.TRUE. /

&DEVC XYZ=2.0,2.0,2.3, QUANTITY='THERMOCOUPLE', ID='TC_ceiling_fire' /
&DEVC XYZ=3.5,3.5,2.3, QUANTITY='THERMOCOUPLE', ID='TC_ceiling_far' /
&DEVC XYZ=3.5,3.5,1.9, QUANTITY='THERMOCOUPLE', ID='TC_z1.9' /
&DEVC XYZ=3.5,3.5,1.5, QUANTITY='THERMOCOUPLE', ID='TC_z1.5' /
&DEVC XYZ=3.5,3.5,1.1, QUANTITY='THERMOCOUPLE', ID='TC_z1.1' /
&DEVC XYZ=3.5,3.5,0.5, QUANTITY='THERMOCOUPLE', ID='TC_z0.5' /

&TAIL /
```

Listing 1: FDS input file for the baseline compartment fire

A.2 Run 2: Energy- and Duration-Matched Fire (5 MW)

```
&HEAD CHID='compartment3', TITLE='Matched fire: 4x4x2.4 m room, 1.0x2.0 m door, 5 MW peak, 8000 MJ total
     , 0.2 m mesh' /

&MESH IJK=30,40,30, XB=-1.0,5.0,-3.0,5.0,0,6.0 /

&TIME T_END=2400.0 /
&DUMP DT_DEVC=1.0, DT_SLCF=10.0 /

&REAC FUEL='PROPANE', SOOT_YIELD=0.01 /

&MATL ID='CONCRETE', CONDUCTIVITY=1.6, SPECIFIC_HEAT=0.88, DENSITY=2300.0 /
&SURF ID='CONCRETE', MATL_ID='CONCRETE', THICKNESS=0.2, COLOR='GRAY' /

&SURF ID='BURNER', HRRPUA=1250.0, RAMP_Q='FIRE', COLOR='RED' /
```

```
&RAMP ID='FIRE', T=0.0, F=0.00 /
&RAMP ID='FIRE', T=60.0, F=0.04 /
&RAMP ID='FIRE', T=120.0, F=0.16 /
&RAMP ID='FIRE', T=180.0, F=0.36 /
&RAMP ID='FIRE', T=240.0, F=0.64 /
&RAMP ID='FIRE', T=300.0, F=1.00 /
&RAMP ID='FIRE', T=1500.0, F=1.00 /
&RAMP ID='FIRE', T=2100.0, F=0.00 /
&RAMP ID='FIRE', T=2400.0, F=0.00 /

&OBST XB=-0.2,4.2,-0.2,0.0,0,2.6, SURF_ID='CONCRETE' /
&OBST XB=-0.2,4.2,4.0,4.2,0,2.6, SURF_ID='CONCRETE' /
&OBST XB=-0.2,0.0,-0.2,4.2,0,2.6, SURF_ID='CONCRETE' /
&OBST XB=4.0,4.2,-0.2,4.2,0,2.6, SURF_ID='CONCRETE' /
&OBST XB=-0.2,4.2,-0.2,4.2,2.4,2.6, SURF_ID='CONCRETE' /

&HOLE XB=1.4,2.4,-0.3,0.1,0,2.0 /

&OBST XB=1.0,3.0,1.5,3.5,0,0.2, SURF_IDS='BURNER','INERT','INERT' /

&VENT MB='XMIN', SURF_ID='OPEN' /
&VENT MB='XMAX', SURF_ID='OPEN' /
&VENT MB='YMIN', SURF_ID='OPEN' /
&VENT MB='YMAX', SURF_ID='OPEN' /
&VENT MB='ZMAX', SURF_ID='OPEN' /

&SLCF PBY=2.5, QUANTITY='TEMPERATURE' /
&SLCF PBX=1.9, QUANTITY='TEMPERATURE' /

&DEVC XYZ=2.0,2.5,2.3, QUANTITY='THERMOCOUPLE', ID='TC_ceiling_fire' /
&DEVC XYZ=3.5,3.5,2.3, QUANTITY='THERMOCOUPLE', ID='TC_ceiling_far' /
&DEVC XYZ=3.5,3.5,1.9, QUANTITY='THERMOCOUPLE', ID='TC_z1.9' /
&DEVC XYZ=3.5,3.5,1.5, QUANTITY='THERMOCOUPLE', ID='TC_z1.5' /
&DEVC XYZ=3.5,3.5,1.1, QUANTITY='THERMOCOUPLE', ID='TC_z1.1' /
&DEVC XYZ=3.5,3.5,0.5, QUANTITY='THERMOCOUPLE', ID='TC_z0.5' /

&DEVC XYZ=2.0,2.5,2.3, ORIENTATION=0,0,1, QUANTITY='ADIABATIC SURFACE TEMPERATURE GAS', ID='
    AST_ceiling_fire' /
&DEVC XYZ=3.5,3.5,2.3, ORIENTATION=0,0,1, QUANTITY='ADIABATIC SURFACE TEMPERATURE GAS', ID='
    AST_ceiling_far' /

&TAIL /
```

Listing 2: FDS input file for the higher-heat-release-rate fire scenario

B Python Implementation

This section contains the Python scripts used to calculate the parametric fire curve, process the FDS results, calculate steel temperatures and compare the resulting thermal responses.

B.1 Parametric Fire and Steel Model

```python


import os
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

LX, LY, H = 4.0, 4.0, 2.4
AV, HEQ = 1.0 * 2.0, 2.0
RHO, C, LAM = 2300.0, 880.0, 1.6
QFD = 511.0
```

```
T_LIM_MIN = 20.0
AM_V = 200.0
ALPHA_PARAM, ALPHA_ISO = 35.0, 25.0

AF = LX * LY
AT = 2 * AF + 2 * (LX + LY) * H
QTD = QFD * AF / AT
O = AV * np.sqrt(HEQ) / AT
b = np.sqrt(RHO * C * LAM)
T_LIM = T_LIM_MIN / 60.0
G_REF = (0.04 / 1160.0) ** 2

print(f"Af={AF:.1f} m2  At={AT:.1f} m2  O={O:.4f} m^0.5 (valid 0.02-0.20: {0.02<=O<=0.20})")
print(f"b={b:.0f} J/m2s^0.5K (valid 100-2200: {100<=b<=2200})  qt,d={QTD:.1f} MJ/m2 (valid
    50-1000: {50<=QTD<=1000})")

gamma = (O / b) ** 2 / G_REF
tmax_vent = 0.2e-3 * QTD / O
fuel_controlled = tmax_vent < T_LIM

def heat(ts):
    return 20 + 1325 * (1 - 0.324 * np.exp(-0.2 * ts) - 0.204 * np.exp(-1.7 * ts) - 0.472 *
    np.exp(-19 * ts))

if not fuel_controlled:
    tmax = tmax_vent
    g_heat = gamma
    tstar_max = tmax * gamma
    x = 1.0
    cool_tstar_max = tstar_max
else:
    o_lim = 0.1e-3 * QTD / T_LIM
    g_heat = (o_lim / b) ** 2 / G_REF
    if O > 0.04 and QTD < 75 and b < 1160:
        g_heat *= 1 + ((O - 0.04) / 0.04) * ((QTD - 75) / 75) * ((1160 - b) / 1160)
    tmax = T_LIM
    tstar_max = tmax * g_heat
    cool_tstar_max = tmax_vent * gamma
    x = T_LIM * gamma / cool_tstar_max

theta_max = heat(tstar_max)
print(f"Gamma={gamma:.3f}  t_max(vent)={tmax_vent*60:.1f} min  {'FUEL' if fuel_controlled
    else 'VENTILATION'}-controlled  "
      f"t_max={tmax*60:.1f} min  theta_max={theta_max:.0f} C")

def parametric(t_s):
    th = t_s / 3600.0
    out = np.zeros_like(th)
    heating = th <= tmax
    out[heating] = heat(th[heating] * g_heat)
    ts = th[~heating] * gamma
    if cool_tstar_max <= 0.5:
        out[~heating] = theta_max - 625 * (ts - cool_tstar_max * x)
    elif cool_tstar_max < 2:
        out[~heating] = theta_max - 250 * (3 - cool_tstar_max) * (ts - cool_tstar_max * x)
    else:
        out[~heating] = theta_max - 250 * (ts - cool_tstar_max * x)
    return np.maximum(out, 20.0)

def iso834(t_s):
    return 20 + 345 * np.log10(8 * t_s / 60.0 + 1)

def ca(m):
```

```
    if m < 600: return 425 + 0.773*m - 1.69e-3*m**2 + 2.22e-6*m**3
    if m < 735: return 666 + 13002/(738 - m)
    if m < 900: return 545 + 17820/(m - 731)
    return 650.0

def steel(t, tg, alpha):
    s = np.zeros_like(t); s[0] = 20.0
    for i in range(1, len(t)):
        g, m = tg[i-1], s[i-1]
        q = alpha*(g-m) + 0.7*5.67e-8*((g+273.15)**4 - (m+273.15)**4)
        s[i] = m + AM_V/(ca(m)*7850.0)*q*(t[i]-t[i-1])
    return s

t = np.arange(0, 3 * 3600 + 1, 1.0)
g_iso, g_par = iso834(t), parametric(t)
s_iso, s_par = steel(t, g_iso, ALPHA_ISO), steel(t, g_par, ALPHA_PARAM)

for mins in (5, 10, 20, 30, 60, 90, 120):
    i = mins * 60
    print(f"{mins:4d} min  gas ISO {g_iso[i]:5.0f} param {g_par[i]:5.0f} | steel ISO {s_iso[i]:5.0f} param {s_par[i]:5.0f}")
print(f"peak steel: ISO(3h) {s_iso.max():.0f} C, parametric {s_par.max():.0f} C at {t[s_par.argmax()]/60:.0f} min")

fig, ax = plt.subplots(1, 2, figsize=(12, 4.5))
ax[0].plot(t/60, g_iso, "k--", label="ISO 834")
ax[0].plot(t/60, g_par, label="EN 1991-1-2 Annex A")
ax[1].plot(t/60, s_iso, "k--", label="steel, ISO 834")
ax[1].plot(t/60, s_par, label="steel, parametric")
ax[0].set_title("Gas temperature"); ax[1].set_title(f"Unprotected steel, Am/V={AM_V:.0f} 1/m")
for a in ax:
    a.set_xlabel("time (min)"); a.set_ylabel("temperature (C)"); a.grid(alpha=0.3); a.legend()
plt.tight_layout(); plt.savefig("parametric_compare.png", dpi=150)
```

Listing 3: Parametric fire curve and steel thermal-response model

B.2 Comparison and Equivalent Exposure Time

```

import csv
import os
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import parametric_curve as pc

AM_V_LIST = (100.0, 200.0, 300.0)
T_END = 3 * 3600.0

KY_T = [20, 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200]
KY_V = [1.0, 1.0, 1.0, 1.0, 1.0, 0.78, 0.47, 0.23, 0.11, 0.06, 0.04, 0.02, 0.0]


def read_devc(path):
    with open(path, newline="") as f:
        rows = [r for r in csv.reader(f) if r]
    names = [n.strip().strip('"') for n in rows[1]]
    data = np.array([[float(x) for x in r] for r in rows[2:]])
```

```
    t, idx = np.unique(data[:, 0], return_index=True)
    return t, {n: data[idx, i] for i, n in enumerate(names) if i > 0}


def steel_am(t, tg, alpha, am_v):
    s = np.zeros_like(t)
    s[0] = 20.0
    for i in range(1, len(t)):
        g, m = tg[i - 1], s[i - 1]
        q = alpha * (g - m) + 0.7 * 5.67e-8 * ((g + 273.15) ** 4 - (m + 273.15) ** 4)
        s[i] = m + am_v / (pc.ca(m) * 7850.0) * q * (t[i] - t[i - 1])
    return s


def t_equivalent(peak, t, s_iso):
    k = np.argmax(s_iso >= peak)
    return None if s_iso[k] < peak else t[k] / 60.0


def ky(theta):
    return float(np.interp(theta, KY_T, KY_V))


t = np.arange(0, T_END + 1, 1.0)
exposures = {
    "ISO 834": (pc.iso834(t), pc.ALPHA_ISO),
    "Parametric (Annex A)": (pc.parametric(t), pc.ALPHA_PARAM),
}

for label, fname in (("compartment1", "compartment1_devc.csv"), ("compartment3", "
    compartment3_devc.csv")):
    if not os.path.exists(fname):
        print(f"[skip] {fname} not found")
        continue
    tf, cols = read_devc(fname)
    for dev in ("TC_ceiling_fire", "TC_ceiling_far"):
        g = np.interp(t, tf, cols[dev], right=cols[dev][-1])
        g[t > tf[-1]] = np.nan
        exposures[f"FDS {label} {dev.replace('TC_ceiling_', '')}"] = (g, pc.ALPHA_PARAM)
    print(f"{fname}: {tf[0]:.0f}-{tf[-1]:.0f} s, {len(tf)} samples, devices: {', '.join(cols)
    }")

results = {}
lines = []
for am in AM_V_LIST:
    lines.append(f"\nAm/V = {am:.0f} 1/m")
    lines.append(f"{'exposure':36s} {'peak gas':>9s} {'peak steel':>11s} {'at (min)':>9s} {'
    t_e (min)':>10s} {'k_y':>6s}")
    s_iso = steel_am(t, exposures["ISO 834"][0], exposures["ISO 834"][1], am)
    for name, (g, alpha) in exposures.items():
        ok = ~np.isnan(g)
        tt, gg = t[ok], g[ok]
        s = steel_am(tt, gg, alpha, am)
        pk, ipk = s.max(), s.argmax()
        te = t_equivalent(pk, t, s_iso) if name != "ISO 834" else None
        results[(am, name)] = (tt, gg, s)
        lines.append(f"{name:36s} {gg.max():9.0f} {pk:11.0f} {tt[ipk]/60:9.1f} "
                     f"{('-' if te is None else f'{te:.1f}'):>10s} {ky(pk):6.2f}")
text = "\n".join(lines)
print(text)
print("\nNOTE: ISO 834 'peak' is the value at the end of the 3 h window, not a real peak.")
open("results_table.txt", "w").write(text + "\n")
```

```
fig, ax = plt.subplots(1, 2, figsize=(12, 4.5))
for name in exposures:
    tt, gg, s = results[(200.0, name)]
    sty = "k--" if name == "ISO 834" else "-"
    ax[0].plot(tt / 60, gg, sty, label=name)
    ax[1].plot(tt / 60, s, sty, label=name)
ax[0].set_title("Gas temperature")
ax[1].set_title("Unprotected steel, Am/V = 200 1/m")
for a in ax:
    a.set_xlim(0, 60)
    a.set_xlabel("time (min)")
    a.set_ylabel("temperature (C)")
    a.grid(alpha=0.3)
ax[0].legend(fontsize=7)
plt.tight_layout()
plt.savefig("steel_comparison.png", dpi=150)
print("wrote steel_comparison.png and results_table.txt")
```

Listing 4: Comparison of gas and steel temperatures across fire scenarios